Hobart Corporation



HLX Scale

Software Architecture Document

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Definitions, Acronyms, and Abbreviation

The following terms and acronyms are used throughout this document.

API	Application Programmer's Interface
Bsquare	Bsquare Corporation
CF	Compact Flash
HOBART_APP	Environment variable holding the path of the *Application* directory
Iso Image	An archive file of an optical disc.
OS	Operating System
PC	Personal Computer
sudo	Ubuntu command used to execute in super user mode

Applicable Documents

The following table contains documents that were referenced during the creation of this document.

Ref. #	Document Name	Document ID/Location
1	Hobart Peripheral Attachment Sequence Diagram.docx	https://externalsvn.camelot.bsquare.com/svn/Hobart/Hobart_Linux_Port/branches/shared_code/Docs/Tech docs
2	HLX OS Paths.xlsx	https://externalsvn.camelot.bsquare.com/svn/Hobart/Hobart_Linux_Port/branches/shared_code/Docs/Tech docs

Table 1 – Applicable Documents

Overview

Based on Hobart's acceptance of the *Linux Distribution Selection Criteria and Recommendation* document, Ubuntu version 9.10 was selected as the Linux platform to be used on the Hobart Linux project. In order to meet Hobart's original requirement for this distribution to fit on a 2GB CF card, Bsquare customized the iso image to reduce the size to approximately 1.6GB. The current Hobart requirement is for the Linux distribution to fit on a 4G CF card. This document details procedure required to setup the Linux distribution on the HLX properly. Additionally, startup tasks initiated for the proper functionality of the HLX suite are listed in this document as well.

Installation

Ubuntu can be installed on a CF or hard-drive directly connected to a HLX. Even though this document details the installation procedure for HLX platform only, a PC may also be used for the procedure. However care must be taken in modifying the Ubuntu bootloader settings to set the installation drive as the primary device.

Hardware required

- USB based External CDROM drive.
- USB cable with *type B* connector on one end and *type A* connecter on the other.
- Custom iso CD.
- 4G CF

Installation procedure

Before the CDROM can be used for installation the HLX BIOS needs to be modified to boot from the CDROM. In order to do this following procedure must be followed:

- Power off the target HLX and disconnect the HLX AC power cable.
- Install a 4GB CompactFlash (CF) card in the HLX. CF label faces toward the rear.
- Attach a USB keyboard and mouse to the HLX.
- Plug-in the USB based external CDROM into the HLX and insert the CD containing Ubuntu custom1.0.iso in it.
- Power on the CDROM and the HLX.
- Enter BIOS setup menu by pressing *Delete* Key during the HLX boot.
- From the BIOS menu select *BIOS CMOS Configuration*.
- Under *BOOT ORDER* set *BOOT 1st* to *CDROM*.
- Press *Esc* key to return to the main menu.
- Choose *Write to CMOS and Exit* option from the menu. When prompted type in 'Y' to confirm changes to the BIOS.

At this point the HLX will continue the boot process and will attempt to boot from the CDROM. Continue with the following steps when HLX has booted the custom iso:

- Choose the language to use for the installation from the menu.
- Press *F4* key on the keyboard. Using the arrow keys highlight *Safe Graphics mode* from the menu and select it by pressing the *Enter* key. A cross 'x' should appear next to it. Press *Esc* key to continue.
- Press *F6* key on the keyboard and highlight *acpi=off.* Press *Enter key* to select it. Highlight *noapic* option from the menu by using the arrow keys. Press *Enter* key to select it. Both these options should now have a cross 'x' beside them. Press *Esc key* to continue.
- Choose *Install Ubnutu* option from the main menu by highlighting it using the arrow keys and pressing the *Enter* key.
- Follow the setup wizard to finish installing Ubuntu.
    - At the "Prepare Disk Space" screen
        - Verify the selected disk is "*SCSI1 (0,0,0) (sda) – 4.0 GB ATA CF Card*" or similar.
        - Choose the option labeled "*Erase and use the entire disk*".
    - At the "Who are you?" screen
        - Enter a unique name for the field *What is the name of this computer?*
        - Choose "*Log in automatically*" option.
    - At the "Ready to Install" screen
        - Click "*Advanced*". The Advanced dialog appears.
        - Confirm "*Install boot loader"* is checked.
        - Under "*Device for boot loader installation*", choose "*/dev/sda*" from the drop list.
        - Click OK. The Advanced dialog clears.
        - Click Install to begin installation.

Once installation is finished, the CDROM must be detached and the BIOS options under *BOOT ORDER, BOOT 1st* must be changed back to *Drive C:* (reference the BIOS setup steps above) before rebooting the HLX.

Linux Startup

Once installation is complete and the HLX reboots, Ubuntu's default boot loader will present boot options. In order to properly manage incompatibility between the Linux kernel and the current version of BIOS (v1.3), kernel boot options must be modified using the following procedure:

- Press *E* key when *GNU GRUB* menu is displayed.
- Move to the end of the 7th line starting with *linux /boot/* and enter *pci=noacpi*.
- Hold down *Ctrl key* and press *X key* to boot Ubuntu.
- After the system boots, open a terminal window by selecting
  Applications->Accessories->Terminal menu option.
- Edit the grub default boot options as follows. A sample file with this change is also located in the repository under config_files/etc/default directory by the name of grub.
  - Type the command *sudo gedit /etc/default/grub* on the terminal window and enter the super user/admin password entered during installation when prompted.
  - Identify the line beginning with GRUB_CMDLINE_LINUX_DEFAULT and add the value " *acpi=noirq*" within the quoted string. Make sure to add the space before the "acpi".
  - Save the file.
  - Type the command *sudo update-grub* on the terminal window.
  - Restart the system by choosing Restart option from the power icon menu.

Setting up Linux environment

Once the Linux installation is complete the Ubuntu environment must be set up properly by installing various packages on the system and by placing various script file in appropriate directories. All required resources are available from the repository: https://externalsvn.bsquare.com/cgi-bin/viewvc.cgi/Hobart/Hobart_Linux_Port/branches/shared_code/.

In order to ensure that all HLX platforms are configured alike the following steps must be taken:

- Install Debian package management frontend, *dselect*, by executing the following command on the terminal window and entering the super user/admin password when prompted

  sudo apt-get install dselect

  sudo command is used to execute a particular command in super user mode.

- Install a preselected set of packages by executing:

  sudo dpkg --set-selections < installed-software.linux.hobart

  Once finished all packages must be updated as follows:
  - Type in *sudo dselect*
  - Choose *Update* from menu
  - Select *Install* to begin the update.

Once all packages have been updated, the required files and folders must be placed in appropriate platform specific directories before the system is ready for operation. On the terminal window type in the following commands in order create appropriate directories and to set their permissions:

sudo mkdir /etc/hobart /lib/firmware/hobart /usr/local/hobart /var/log/hobart ~/hlx ~/.config/autostart

chmod 777 /etc/hobart /lib/firmware/hobart /usr/local/hobart /var/log/hobart /var/www ~/hlx ~/.config/autostart

mkdir /usr/local/hobart/system /usr/local/hobart/user

mkdir /usr/local/hobart/system/fonts /usr/local/hobart/system/flashkeygfx /usr/local/hobart/system/graphics /usr/local/hobart/user/fonts /usr/local/hobart/user/graphics

Once the system has all the required directories, the following table can be used to place various scripts, files and folders in their appropriate location:

Folder/File Name	Linux Path
Fstab	/etc
rc.local	/etc
vsftpd.conf	/etc
Dbempty	/etc/hobart
Dbfiles	/etc/hobart
Dbmirror	/etc/hobart
Text.en_CA.txt	/etc/hobart
Text.en_US.txt	/etc/hobart
Text.es_MX.txt	/etc/hobart
version.ini	/etc/hobart
Xdefaults.en	/etc/hobart
Xdefaults.es	/etc/hobart
defaults_linux.ini	/etc/hobart
bootParams.cfg	/etc/hobart
config_linux.dat	/etc/hobart
Grub	/etc/default
10-local-rules	/etc/udev/rules.d
Daemon (chobart)	/usr/bin
webs.exe	/usr/bin
win_cmd.sh	/usr/bin
www_cmd.sh	/usr/bin
xvkbd.sh	/usr/bin
System	/usr/local/hobart/system
System Fonts	/usr/local/hobart/system/fonts
Flashkey Graphics	/usr/local/hobart/system/flashkeygfx
System Graphics	/usr/local/hobart/system/graphics
User	/usr/local/hobart/user
User Fonts	/usr/local/hobart/user/fonts
User Graphics	/usr/local/hobart/user/graphics
WebPages	/var/www
debug.out	/var/log/hobart
vmTmp	/tmp
Firmware	/lib/firmware/hobart
startup_firefox	home[1]
firefox-3.5.desktop	home/.config/autostart
maximus.xml	home/hlx

Figure 1 Linux Directory Layout

1 Base directory of the logged in user.

Once all files have been copied to appropriate directories, the gconfig-editor utility must be used to disable the folder window from being displayed to the user at auto-mount of a thumb drive. This step is

important because users gain access to the *Service mode* based on the presence of appropriate files on the thumb drive. While it is desirable to auto-mount the thumb drive, allowing access to the contents of the thumb drive poses a security risk. Following steps must be followed to disable this feature:

- Type *gconf-editor* and press *Enter key to start GNOME Configuration Editor*.
- Browse to *apps->nautilus->preferences* using the tree control in the pane located on the left hand side of the *Configuration Editor – preferences* window. Locate the *media_automount_open* option from the choices presented in the right side pane of the *Configuration Editor – preferences* window. Uncheck this option to prevent folder window from popping up at auto-mount.

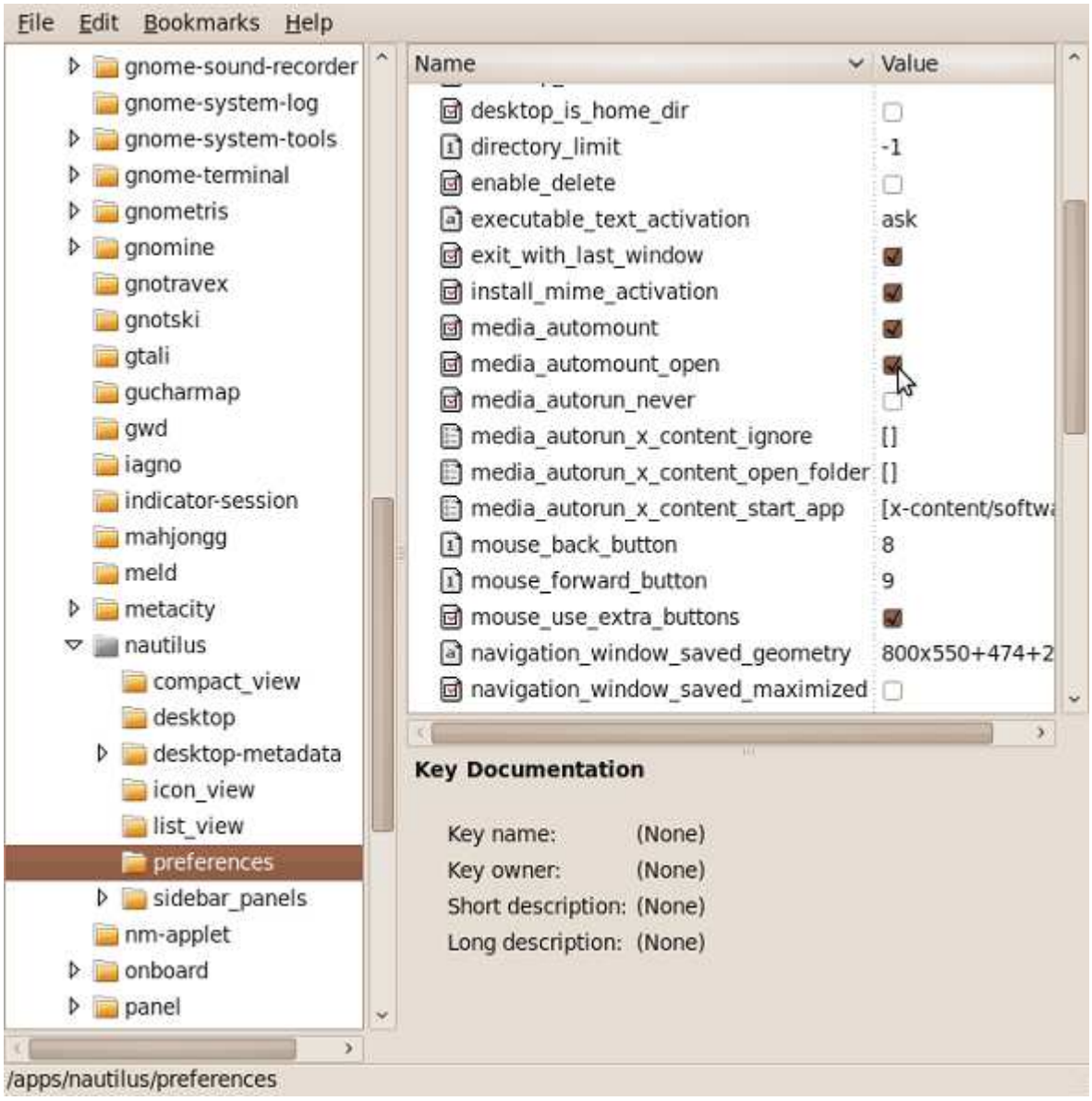


Figure 2.2 Disabling auto-mount folder window

Touch screen driver setup

In order to operate the HLX without a keyboard and mouse, the touch screen driver must be installed. Following steps are required to install and properly calibrate the touch screen driver on the HLX:

- Copy *Tsharc305BETA.tar.gz* to the HLX and uncompress it using the following steps:

  mkdir ~/tsharc
  cd ~/tsharc
  tar -xvzpf Tsharc305BETA.tar.gz

- Enter the Tsharc305BETA folder and run installation script:

  cd Tsharc305BETA
  sudo ./install.sh

- Type *2* to select *Toggle TSHARC SERIAL controller on COM1 (/dev/ttyS0)* menu option.
- Type *12* from the menu next to *Install selected components.*
- The installation script should automatically exit at this point. Type in menu option *13* in order to exit manually.
- Begin the calibration process by typing

  hlincal

- Select menu option *3* (*Calibration*) from the menu that is displayed.
- Choose *9 point calibration* by selecting menu option *3*.
- Begin calibration by selecting menu option *6 (Begin GUI Calibration)*
- Touch the center of each target that is displayed on the screen for about a second.
- Drag your finger around on the test screen that is presented during the 9 point calibration to see if the mouse cursor follows it correctly.
- Press the "Accept" button if calibration is successful.
- Select *P* to return to previous menu.
- Select *O*, *OK*- to save current data and exit.
- Reboot the HLX by choosing *Restart* menu option from the power icon menu.

Appendix A

Startup Tasks

Following table lists the tasks specific to this document which are executed at startup:

Tsharc Touch Screen Daemon	boot.tsharc
Web Server	webs
Hobart Daemon	chobart_d